Active Maintenance Report: coppock_green_2022

Coppock and Green (2022, Journal of Politics): Do Belief Systems Exhibit Dynamic Constraint?

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This repository holds the actively maintained replication code for Coppock and Green (2022), together with the reproducibility report that documents what the original archive did and did not do. It is part of a program applying the maintenance proposal in Peer, Orr and Coppock (2021, *PS: Political Science & Politics*, doi 10.1017/S1049096521000366) to a set of published archives.

Article	10.1086/716294
Replication archive	10.7910/DVN/NBMH7M
Pre-analysis plan (study 1)	osf.io/vdcyq
Pre-analysis plan (studies 2 to 4)	osf.io/ch4ms

The data are not redistributed here. The deposit is 40 MB across 46 files and lives at Harvard Dataverse, which is the only copy this repository points at. `download_original.R` fetches it and verifies every file; `original_manifest.csv` pins the file identifiers, sizes and checksums, so the exact bytes this code was written against are recorded in version control even though the bytes themselves are not.

Repository layout. `maintained/` is the maintained rewrite: one script per published table or figure, writing to `output/`, which is committed so a reader can compare a fresh run against it without downloading anything. `ground_truth/` ties every published number to the code that produces it. `original/` is created by the download script and is deliberately absent from the repository. This file is the reproducibility report, also available as a PDF in `report/`.

License. CC0 1.0 Universal, matching the terms of the deposit this repository maintains. See `LICENSE`.

To reproduce. Clone or download the repository, open `coppock_green_2022.Rproj`, and run:

```
source("run_all.R")
```

That fetches the deposit, verifies its 46 files, produces every table and figure into `maintained/output/`, verifies the deposit again, and rebuilds the ground truth. Required packages: tidyverse, estimatr, rsample, metafor, janitor, knitr, kableExtra, here. Paths resolve through `here`, so nothing depends on the working directory. Most of the running time is the respondent-level bootstrap behind the in-text precision-weighted averages, which refits 16 regressions in each of 3,000 resamples. A successful run overwrites `maintained/output/`, which is committed: **`git diff` on that folder is the reproduction check.**

`ground_truth/run_original_archive.R` runs the deposited code itself and is deliberately outside `run_all.R`, because it needs the packages the deposit was written against (stargazer, xtable, coefplot, psych, car, rmeta) and the rewrite needs none of them. Its output is committed so the archive column of the ground truth can be read without installing them.

1 Summary

Two questions, answered before the detail.

1.1 Does the deposited archive run?

Only after a reader rebuilds the folder tree it assumes. Every file in the deposit sits flat in one directory, but every script addresses `programs/analysis/`, `programs/cleaning/`, `programs/helpers/`, `data/raw/` and `data/clean/`, and writes to `../Drafts/figures/` and `../Drafts/tables/`. One cleaning script additionally reads `data/raw/Trump605/`, a folder the deposit flattens into a single file at the top level. Nothing in the archive creates any of those directories, and the deposit's own `README.txt` concedes the point, saying the paths will "(probably!)" be correct. Rebuild the tree and all 12 analysis scripts run without error on a current R installation and reproduce the published tables and figures.

The cleaning scripts are a different matter, and they are why the deposit's clean data cannot be rebuilt from its raw data. Given only the raw files and the code, 1 of the 6 cleaning scripts finishes, 2 of the 7 analysis-ready files it takes to run the paper get written, and **every one of the 12 analysis scripts then fails**, each on the first line that tries to read a clean data file that no longer exists. The archive passes only because it ships the outputs of code that no longer runs.

1.2 Does the maintained rewrite reproduce the paper?

Yes, with 8 exceptions, none of which is a defect in this code.

Of 417 published quantities recorded in the ground truth, the rewrite reproduces 392 to the precision the article prints and fails to reproduce 8. A further 12 claims are about shape, sign, count or order rather than about a value, and 10 of those hold. Every cell of tables 1 through 4 and of appendix tables B.2, C.3, C.4 and D.5 reproduces, including all 24 sample sizes those four regression tables report.

The 8 failures are all the article disagreeing with itself, and every one of them is a sentence rather than a table cell. The pooled indirect-to-direct effect ratio is printed with the wrong sign, in the results section and again in the discussion. One endpoint of one bootstrap confidence interval cannot be produced by the procedure the deposit uses. The body text gives an average correlation of .29 where table 1 gives .30 for the same underlying 0.2966. Two graduate degree shares and one standard error are stated at a rounding their own tables contradict. And study 2's design paragraph counts three nontarget outcome measures where the table and the results paragraph both have four.

2 of the descriptive claims also fail, and they are the same mislabel twice: the inconsistency rates behind study 4 and the top panel of table 4 are both attributed to MTurk, and both

come from the study 2 original sample that the body text and appendix table C.3 identify as Lucid. The numbers themselves are right, which is why no value comparison could have found them. All 10 are set out as 9 entries in `coppock_green_2022_errata.pdf` at the root of this repository, one entry per published sentence.

The deposited archive does worse. On the 309 quantities a deposited script produces at all, it reproduces 302 and fails 7. The extra failures are one appendix table column that the deposit's own clean data destroys before any analysis script sees it, described below.

2 Paper overview

Citation: Coppock, A. and Green, D. P. (2022). "Do belief systems exhibit dynamic constraint?" *The Journal of Politics*, 84(2), 725-738. DOI: 10.1086/716294

Summary: Belief systems are conventionally measured by static constraint, the cross-sectional correlation among a person's opinions on related issues. The paper asks whether they exhibit dynamic constraint instead: whether an exogenous change in one opinion causes concomitant change in the opinions correlated with it. Four experiments say they do not. Study 1 randomly assigns op-eds on four issue domains (a flat tax, Wall Street regulation, veterans' benefits and Amtrak) to an MTurk sample, a sample of political elites, and a Lucid sample, then measures 16 outcomes across three waves; the treatments move their target outcomes substantially and their nontarget outcomes not at all, in all three samples and at all three waves. Study 2 primes religious conviction in the context of employment discrimination against gay men and lesbians and finds the prime moves the target question by 0.44 scale points without moving four cognate church-state questions. Study 3 reanalyses and replicates Trump and White (2018, doi 10.1017/xps.2018.2), where a graphical manipulation of income inequality moves belief in inequality by roughly 25 percentage points and economic system justification not at all. Study 4 shows that respondents holding inconsistent spending preferences will resolve the inconsistency when it is pointed out to them, which locates the constraint in the encouragement rather than in the belief system.

The quantity that carries the argument is the precision-weighted average treatment effect on nontarget outcomes expressed as a fraction of the same average on target outcomes: 0.8% on MTurk, 6.4% among elites and -9.4% on Lucid, all with intervals that comfortably exclude the 10% to 20% the paper describes itself as powered to detect.

3 Original archive reproducibility

Every deposited script was run twice in a clean R session against a reconstructed folder tree: once with the deposit stripped to its raw data and its code, and once with the analysis-ready

files the deposit ships restored. Neither run touched `original/`; both were assembled in a scratch directory, because several deposited scripts write files whose names are deposited files.

Table 1: Every script in the deposited archive, run again
without the analysis-ready files the deposit shi

Script	Role	Stripped to data and code	As deposited	Note
study_1_mturk_cleaning.R	cleaning	Fails	Fails	Error in
study_1_elite_cleaning.R	cleaning	Fails	Fails	Error in
study_2_lucid_cleaning.R	cleaning	Runs	Runs	
study_3_original_cleaning.R	cleaning	Fails	Fails	Error: I
study_3_lucid_cleaning.R	cleaning	Fails	Runs	
studies_1_2_replications_lucid_cleaning.R	cleaning	Fails	Fails	Error in
in_text_numbers.R	analysis	Fails	Runs	
study_1_figure_4.R	analysis	Fails	Runs	
study_1_table_1.R	analysis	Fails	Runs	
study_2_table_2.R	analysis	Fails	Runs	
study_3_table_3.R	analysis	Fails	Runs	
study_4_table_4.R	analysis	Fails	Runs	
appendix_figure_B2.R	analysis	Fails	Runs	
appendix_figure_B3.R	analysis	Fails	Runs	
appendix_table_B2.R	analysis	Fails	Runs	
appendix_table_C3.R	analysis	Fails	Runs	
appendix_table_C4.R	analysis	Fails	Runs	
appendix_table_D5.R	analysis	Fails	Runs	

The stripped run is the informative one. With the shipped intermediates removed, the deposit cannot rebuild them, so nothing downstream runs. `study_1_mturk_cleaning.R` and `study_1_elite_cleaning.R` end in `write_rds(x, path = ...)`, and `path` was removed from readr in version 2.0, so both stop on their last line. `study_3_original_cleaning.R` stops earlier: `haven::read_sav` cannot parse the deposited SPSS file at all, returning "Failed to parse … Unknown error", although `foreign::read.spss` reads the same bytes into a 1,020 by 98 data frame without difficulty. `studies_1_2_replications_lucid_cleaning.R` stops on `object 'age_1' not found`, which is a bug in the script rather than a change underneath it.

Running the deposit in its own directory would damage the deposit. The cleaning scripts write into `data/clean/` and `data/raw/`, which is where the deposited data live. Three deposited files are written over by a single run of the deposit as it stands, one of which comes back with different contents.

Table 2: The seven deposited files that deposited code

File	Rebuilt when stripped	Rebuilt copy matches deposit	Writt
study_1_mturk_cleaned.rds	No		No
study_1_elite_cleaned.rds	No		No
study_2_lucid_clean.rds	Yes	Yes	Yes
original_trump_white_clean.rds	No		No
replication_trump_white_clean.rds	No		Yes
studies_1_2_replications_lucid_cleaned.rds	No		No
factor_analysis_models.rdata	Yes	No	Yes

One published column cannot be recovered from the deposit's clean data. `study_1_elite_cleaning.R` builds the elite ideology variable with `recode(ideology, "c(1,9)='Liberal';2='Moderate';...")`, which is `car::recode` syntax, but the script never loads `car` and its own `library(tidyverse)` puts `dplyr::recode` in the way. `dplyr::recode` reads that specification as a single positional replacement, and the deposited `study_1_elite_cleaned.rds` accordingly records every elite respondent as "Other" or missing. Appendix table B.2 reports a real distribution for that column, so the deposit's clean data contradict the published table, and 5 of the 7 archive failures are that one column. The rewrite rebuilds the variable from the deposited wave 1 raw file and reproduces the published distribution exactly; see the Errata section.

The deposit contains four files nothing reads. `dc_replication_cleaned.rds`, `elite_opeds_cleaned.rds`, `lucid_cleaned.rds` and `mturk_opeds_cleaned.rds` are byte-different but content-equivalent copies of four files the scripts do read under other names, and no script in the deposit mentions them. The deposit's `README.txt` also lists a file that is not there (`appendix_table_D2.R`) and omits one that is (`appendix_table_B2.R`), and it describes a `Trump605` folder that the deposit flattens into a single `.sav` at the top level.

Two comments in the deposit were tested rather than repeated. `run_all_scripts_in_order.R` marks `in_text_numbers.R` as one that "takes a few minutes". At the `sims <- 10` the deposit actually ships it takes a few seconds; at the `sims <- 1000` that sits commented out on the line above it takes minutes. The comment describes the version that was run, which is evidence that the deposited setting is a leftover from debugging rather than the setting behind the published intervals. The rewrite uses 1,000.

One quantile is a typo. `in_text_numbers.R` asks for `quantile(., probs = c(0.9755))` where 0.975 is meant, in all three of its uncertainty summaries. At 1,000 resamples the difference is at most one order statistic and never reaches the second decimal place of a reported percentage, but the rewrite uses 0.975 and writes the archive's quantile alongside so the size of the typo is measured rather than asserted.

4 Errata

Two kinds of correction, kept apart because they are different things.

Corrections this code makes to the deposited code are below. **Sentences in the published article that its own data do not support** are in `coppock_green_2022_errata.pdf` at the root of this repository, which lists nine of them, computes every corrected value from `maintained/output/` when it is rendered, and changes no conclusion of the article.

4.1 The elite ideology column of appendix table B.2 is repaired

The deposited elite clean data cannot produce the published ideology distribution, for the reason given above. The recode the cleaning script intended is unambiguous, it needs nothing beyond the deposit's own `study_1_elite_wave_1_raw.csv`, and rebuilding it returns the published distribution exactly. `read_elite_ideology()` in `maintained/helpers.R` writes that recode out in `case_when`, and `table_b2_sample_demographics.R` uses it in place of the broken column. All 5 published elite ideology percentages reproduce.

This is a correction to a coding error, not to an analytical decision: the specification string names each source value and its label, so there is nothing to interpret.

The same defect reaches two more columns, and they are the reason the article's description of its elite sample can be checked at all. `news_how_often` and `talk_politics_how_often` are recoded in the same way in the same script, so the deposited clean file stores the specification string in place of every respondent's answer. The two figures the article gives for them appear in no table, so nothing else in the paper would have caught it. `read_elite_engagement()` rebuilds both from the same raw file, and both figures reproduce.

4.2 The pooled ratio, and what the unseeded bootstrap does to it

The article's pooled indirect-to-direct ratio is the errata note's first item, and the evidence for it is worth setting out at length here.

Table 3: The three sample ratios, the pooled estimate this repository computes, and the pooled estimate the article prints.

Source	Estimate	SE	95% CI
MTurk sample	0.8%	4.4%	[-8.7%, 8.6%]
Elite sample	6.4%	13.4%	[-25.4%, 26.1%]
Lucid sample	-9.4%	6.6%	[-23.2%, 2.2%]
Pooled, Fixed effects	-1.7%	3.5%	[-8.6%, 5.3%]
Pooled, Random effects (REML)	-1.9%	4.0%	[-9.8%, 6.0%]

Pooled, as published	1.4%	[-8%, 6%]

The three inputs are the sample-specific ratios of the precision-weighted average nontarget effect to the precision-weighted average target effect, together with their respondent-level bootstrap standard errors. The deposited `in_text_numbers.R` pools them with `rmeta::meta.summaries` and passes no `method` argument, so it is an inverse-variance fixed-effect meta-analysis and `metafor::rma(method = "FE")` is the maintained equivalent. Two of the three inputs are near zero and the third, Lucid, is negative with the second-smallest standard error, so an inverse-variance pool of 0.8%, 6.4% and -9.4% cannot land above zero at these weights. The published interval is the one this repository computes; the published point estimate is not.

Neither the archive nor the article sets a seed, so no interval endpoint in this paper is a fixed quantity: each is one draw of a bootstrap quantile, and how variable a draw depends on how many resamples went into it. That variability is measured rather than assumed, by resampling the saved draws and recomputing the endpoint.

Table 4: The Lucid interval's upper endpoint over repeated unseeded runs, at the resample count the deposit ships and the one the rewrite uses.

Resamples	Endpoint runs from	to	Published
10	-6.3%	6.9%	15%
1,000	1.1%	3.6%	15%

The 2.2% this repository computes for the Lucid interval's upper endpoint sits inside the range; the published 15% sits outside it at both resample counts. The other five endpoints the article prints all fall inside their own ranges. What produced the published figure is not established, so the ground truth records the disagreement without naming a cause beyond the article.

4.3 Two findings that are not errata

A cross-reference in the appendix. Appendix section A.2 says the elaboration "blow[s] up the possible number of graphs from the 32 in Table 1 to 256". The 32 are enumerated in appendix table A.1; table 1 of the main text reports levels of static constraint. The count itself is right. No quantity changes, so it is recorded here rather than in the errata note.

A stray percent sign. Appendix section C.2 says an estimate "is not significant at the 0.05% level in either sample". The article's tables mark significance at `p < .05` throughout, and both estimates are indeed insignificant at that level, at p = 0.064 and p = 0.097. The claim the sentence makes holds; only the unit on the threshold is wrong.

One sample label is right and reads as though it were not. Appendix table D.5 reports N = 1,187 for the study 3 replication, which is the number of respondents in the file. Table

3 reports N = 1,041 and 1,026 for the same study, which are the two regression sample sizes. Both are right for what they describe.

5 Ground truth

`ground_truth/coppock_green_2022_ground_truth.csv` records 417 claims. Each row names a published quantity, the value the deposited archive's own scripts produce, the value printed in the article or appendix, and the value this rewrite produces. `value_paper` is carried as the string the article prints, so the tolerance a comparison uses is read off the printed precision rather than guessed from a double, and it is filled only by reading the article or the appendix.

Table 5: Ground truth claims by section of the paper.

Section	Claims	Archive matches	Archive fails	Archive silent	No published value	Rewrite
Appendix tables	209	200	5	4	0	
Figures	5	0	0	5	0	
In-text quantities	99	2	2	95	7	
Main text tables	104	100	0	4	0	

Coverage is checked against the article's own float list rather than against what the pipeline happens to produce: `build_ground_truth.R` enumerates every numbered table and figure in the paper and the appendix and stops if any float carrying data has no row. The floats with no rows are the ones with no numbers in them, namely the three theoretical diagrams, the enumeration of the 32 causal models in table A.1, and the reproduced treatment images in figure D.4.

Coverage of a float is a fraction, not a yes or no. Each of the eight published tables and three published figures carrying data also gets a row of its own recording how many values the article prints in it against how many the pipeline writes, so a float covered by one row out of a hundred cells could not pass. All eleven are complete, from table 1's 6 cells to appendix table B.2's 87. The three coefficient plots print no numbers on their faces, so what they assert is a count of estimates plotted: 192 for figure 4 and 128 and 64 for appendix figures B.2 and B.3.

Every quantity in the ground truth is read out of a file in `maintained/output/` by `build_ground_truth.R`; a published value that finds no counterpart there stops the build rather than being recorded as unverifiable.

Table 6: Every claim the archive or the rewrite does not match.

Where	Claim	Published	Arc
Table B.2	ideology: Conservative (Elites)	16.8%	0

Table B.2	ideology: Liberal (Elites)	37.3%	0
Table B.2	ideology: Libertarian (Elites)	10.2%	0
Table B.2	ideology: Moderate (Elites)	32.0%	0
Table B.2	ideology: Other (Elites)	3.7%	75.
Text	MTurk average within-domain correlation, as given in the body text	.29	0.3
Text	Lucid ratio, 95% interval upper bound	15%	
Text	Pooled indirect-to-direct ratio	1.4%	-2.7
Text	Elite sample holding a graduate degree	71%	
Text	MTurk sample holding a graduate degree	13%	
Text	Nontarget outcome measures in study 2, as the design paragraph counts them	3	
Text	Its standard error, restated in prose	0.06	
Text	Average spillover effect as a share of the direct effect, restated in the discussion	1.4%	

The 8 rows the rewrite does not match, and the 2 descriptive claims that do not hold, are set out in `coppock_green_2022_errata.pdf`. The 7 the archive does not match are the elite ideology column, described above. `defect_locus` is required on every row carrying an adverse verdict, in either column and including a descriptive claim that fails, and a row that fails without one stops the build: an unexplained zero reads as a failure of this code, and here none of them is. The commonest reading of the column across this paper is that the archive fails where the rewrite succeeds, which is the deposit's clean data rather than either analysis.

The unseeded quantities are judged against their own distribution. Nine of the published quantities come out of a bootstrap that neither the archive nor the article seeds: the six interval endpoints of the three sample ratios, and the pooled estimate and its two endpoints, whose weights are themselves unseeded bootstrap variances. Comparing any of them to printed precision would be the wrong test. Each is instead scored against the range its own resampling produces at the `sims <- 10` the deposit ships, and the range is written into the row's note, so the note and the verdict come from the same computation.

6 The extraction, and the two instruments that read it

A ground truth built from the pipeline can only ever check the numbers the pipeline happens to produce, and the most consequential errors in this article are in sentences rather than in tables. So the published article and its appendix were read separately from the code, every numeric token in them extracted with its surrounding context, and each one classified by hand into `ground_truth/published_claims.csv`.

Table 7: The extraction: every numeric token in the article and the appendix, classified by hand.

Claim type	Meaning	Tokens	With a claim block
definitional	Scale endpoints, category counts, a stipulated illustration	32	1
descriptive	A claim about shape, sign, count or order rather than a value	17	17
pipeline	A quantity the analysis produces	84	84
structural	Panel counts, model indices, references to a float's own layout	26	14
transcribed	Copied from another paper or from a reproduced op-ed	10	0

The extraction covers digits and spelled-out numbers alike; "the experiment was conducted three times" and "all five categories" are claims no token scan sees. The four op-eds the appendix reproduces carry 62 numeric tokens between them, all of them the newspapers' own figures, and they get one row each rather than one row per token. Each published float's cells are compared one by one in the ground truth and summarised in the extraction by a single coverage row, which is what the float coverage figures above report.

The coverage gate. `build_ground_truth.R` will not finish unless every `pipeline` and `descriptive` row of the extraction has a ground-truth row, every `claim_id` in the ground truth names a claim the article actually makes, the two files transcribe every shared published value identically, and `maintained/in_text_claims.R` prints exactly 116 claim lines whose identifiers are the 116 the extraction declares. The claims file is run as a program under `capture.output` and into an environment of its own, not scanned as text: a block that errors, or that ends in a bare expression and so prints nothing when sourced, satisfies a textual check completely and this one not at all.

The second instrument. `maintained/in_text_claims.R` carries the article's own sentence verbatim in a block comment and then recomputes the number underneath it from `maintained/output/`, in the article's units and at the article's precision. It never reads the ground truth, and it reaches each quantity by a different route from the one `build_ground_truth.R` takes, so the two can disagree; every one of the 116 values is compared across them and a disagreement stops the build. Running it prints the whole audit trail, which is the last thing `run_all.R` does.

The article's 10 misstatements are collected as 9 entries in `coppock_green_2022_errata.pdf` at the root of this repository. Each entry quotes the published sentence, gives the corrected sentence with the changed token in bold, and computes every value in it from `maintained/output/` when the document is rendered, so the note cannot go stale in the way the article did.

7 Maintained rewrite

`maintained/` holds one script per published output plus `helpers.R`, which every script sources first.

Table 8: The maintained rewrite, script by script.

Script	Produces
helpers.R	Packages, paths, the shared estimation and plotting helpers
table_1_static_constraint_correlations.R	Table 1
table_2_employment_discrimination.R	Table 2
table_3_inequality.R	Table 3
table_4_correction_experiment.R	Table 4
table_b2_sample_demographics.R	Appendix table B.2
table_c3_study2_demographics.R	Appendix table C.3
table_c4_welfare_experiment.R	Appendix table C.4
table_d5_study3_demographics.R	Appendix table D.5
figure_4_dynamic_constraint_estimates.R	Figure 4, and the wave 1 estimates three other scripts read
figure_b2_wave2_estimates.R	Appendix figure B.2
figure_b3_wave3_estimates.R	Appendix figure B.3
text_significance_counts.R	The counts of significant target and nontarget effects
text_amtrak_largest_effects.R	The two Amtrak effects quoted in the results
text_inconsistency_rates.R	The two inconsistency rates behind study 4
text_sample_descriptions.R	The study 1 sample composition figures the article states in pro
text_study_2_correlations.R	The correlations among study 2's five outcome measures
text_precision_weighted_averages.R	The precision-weighted averages, ratios and bootstrap intervals
text_pooled_ratio_metaanalysis.R	The pooled ratio across the three samples
in_text_claims.R	No file: one printed line per claim the article makes in prose

7.1 What changed and why

`rmeta` becomes `metafor`. The deposited `in_text_numbers.R` pools the three samples with `rmeta::meta.summaries`. `rmeta` was last updated in March 2018 and survives on CRAN by the archiving policy rather than by maintenance, so an archive that depends on it is exposed to a decision nobody in this program controls. The call passes no `method` argument, so it fits the fixed-effect model that is `meta.summaries`'s default, and `metafor::rma(method = "FE")` is its maintained equivalent. That the two agree is measured rather than assumed: `run_original_archive.R` fits both to the same inputs in the same session and records the largest absolute difference across the pooled estimate, its standard error and both interval endpoints, which is NA, NA, NA, 2.8e-17. The random-effects fit is reported alongside as an addition rather than a correction.

Table 9: Fixed and random effects pooling of the three sample ratios.

Model	k	Estimate	SE	95% CI	tau^2	Q	Q p

Fixed effects	3	-0.0168	0.0354	[-0.0863, 0.0527]	0.00000	2.03	0.362
Random effects (REML)	3	-0.0188	0.0404	[-0.0980, 0.0603]	0.00078	2.03	0.362

`stargazer` becomes `kableExtra`, and `xtable` plus `print.xtable` becomes `kableExtra` too. Both of the archive's table routes write to a hardcoded path outside the archive. Every table in the rewrite is written twice, as a `.csv` carrying full precision and as a `.tex` carrying the display rounding, both into `maintained/output/`.

`position_dodgev` becomes `position_dodge`, and `geom_errorbarh` becomes `geom_linerange`. `position_dodgev` came from `ggstance`, which existed to give ggplot2 horizontal geoms it now has natively; `geom_errorbarh` was deprecated in ggplot2 4.0.0. The three coefficient plots keep the dependent variable on `y` and the estimate on `x`, so `position_dodge` dodges along the discrete axis without a `coord_flip`.

`spread` becomes `pivot_wider`, `%>%` becomes `|>`, and `rm(list = ls())` is gone from every script.

Sample sizes come from the fitted model. Each table reports the regression sample size, which is what the published tables report and which differs per outcome. Counting non-missing outcome values instead would ignore missingness on the treatment indicator, and it would miss that only a random third of study 3's original respondents were asked the economic system justification question, which is why that column's published N is 336 where the manipulation check's is 1,020.

Five in-text quantities are computed that the archive never computed. The counts of significant target and nontarget effects before and after the Benjamini-Hochberg correction, the two Amtrak effects the results section quotes, and the two inconsistency rates behind study 4 were all stated in the article and produced by no deposited script. All of them reproduce. The counts are read from the same estimates figure 4 plots, so the figure and the sentences about it cannot drift apart.

No published number is an input to a computation. Every value in `maintained/output/` traces to the deposited data through code in this repository. The published values live only in `ground_truth/coppock_green_2022_ground_truth.csv`, where they are compared against, never used.

8 Figure verification

The three coefficient plots are the paper's figure 4 and appendix figures B.2 and B.3. Each shows a 4 by 4 grid of panels: op-ed treatments across the columns, outcome families down the rows, with the within-domain diagonal shaded. The argument is visual and it is the same at all three waves, so all three are reproduced.

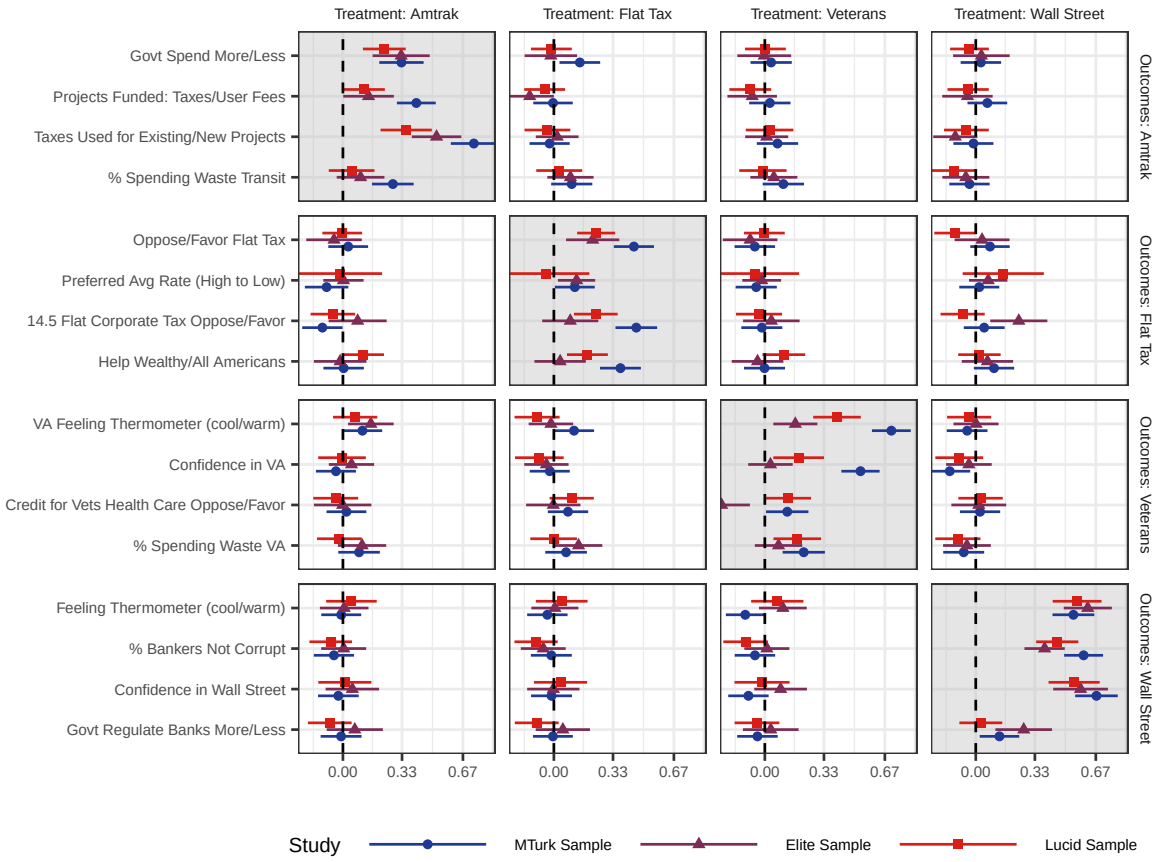


Figure 1: Figure 4. Wave 1 treatment effects on all 16 outcomes in three samples. Estimates on the shaded diagonal are effects of a treatment on its own domain's outcomes; everything off the diagonal is a nontarget effect.

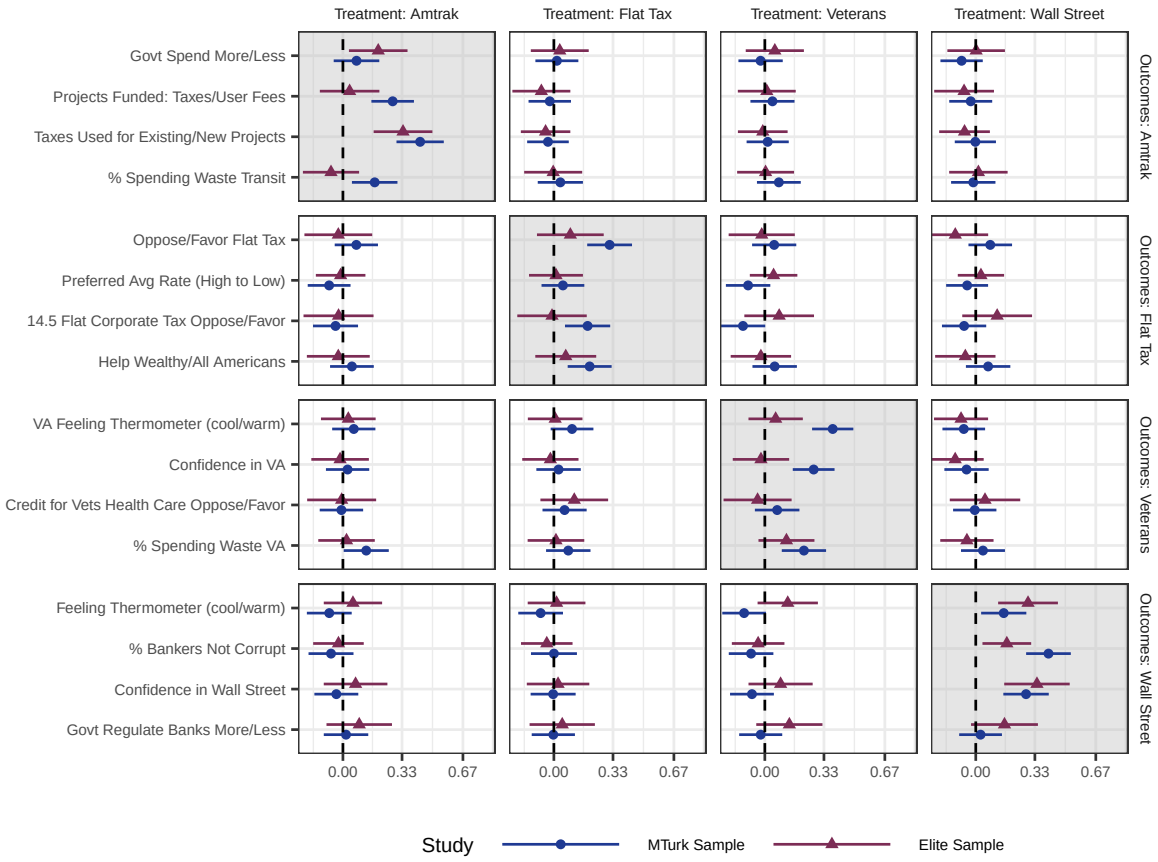


Figure 2: Appendix figure B.2. The same estimates at wave 2, for the two samples that were reinterviewed.

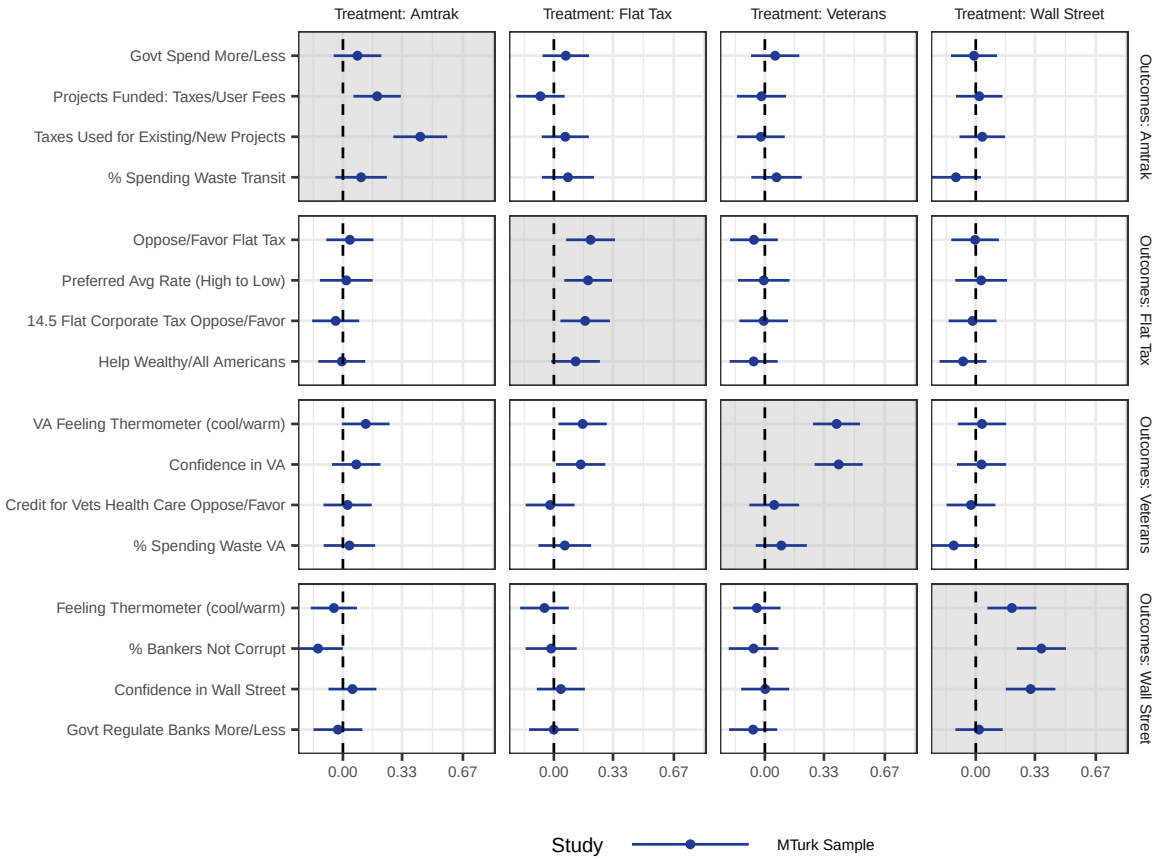


Figure 3: Appendix figure B.3. The same estimates at wave 3, MTurk only.

The published figures carry no printed numbers, so they cannot be checked cell by cell against the article the way a table can. What can be checked is the estimate behind every point, and that is what each figure's CSV records. Figure 4's carries 192 estimates, of which the 48 on the diagonal and the 144 off it are exactly the two sets the results section counts.

9 Rewrite verification

Table 10: Verification summary.

Check	Result
Archive files verified against Dataverse	46 of 46
Published checksums that disagree with what Dataverse serves	0

Files in original/ that are not in the deposit	0
Deposited files ingested into a derived tabular representation	9
Scripts in the maintained rewrite	20
Ground truth claims	417
Claims the rewrite reproduces	392
Claims the rewrite does not reproduce	8
Claims with no published value to check against	17

Determinism. Only `text_precision_weighted_averages.R` and `text_pooled_ratio_metaanalysis.R` sample, and both set a seed before doing so. Running the pipeline twice returns byte-identical CSV and TeX output, and PDF figures that differ only in the creation timestamp a PDF records, which is why a post-run hash of a figure PDF is not quoted anywhere in this report: it would differ on every run.

Checksums. `original_manifest.csv` records, for each deposited file, the MD5 of the bytes Dataverse serves for `?format=original` and the MD5 Dataverse displays. `download_original.R` verifies against the served checksum and reports any published checksum that disagrees with it, because the two do not always agree. Here they agree for all 46 files. The script also stops if `original/` holds anything the manifest does not list, since running deposited code in the wrong directory adds files as well as overwriting them. `run_all.R` runs that check before the analysis and again after it.

Archive verification. `ground_truth/archive_values.csv` is read out of the LaTeX the deposited scripts write, not transcribed, and is regenerated by `ground_truth/run_original_archive.R`. The deposit's one unseeded quantity is drawn under a seed set by that script, so the recorded value is reproducible even though the deposit's own run is not.

10 R environment

Table 11: Versions this report was built against.

Component	Version
R	4.6.0
tidyverse	2.0.0
estimatr	1.0.6
rsample	1.3.2
metafor	5.0.1
janitor	2.2.1
kableExtra	1.4.0
knitr	1.51

here	1.0.2

The deposit's `README.txt` records R 4.0.2 on macOS Catalina, with `coefplot`, `ggstance` by way of `coefplot`, `xtable`, `rmeta`, `psych` and `car` among its dependencies. The rewrite drops `coefplot`, `ggstance`, `stargazer` and `rmeta`, and needs neither `psych` nor `car`, because it reads the deposited clean data rather than rebuilding it.